Luvian

AI-Native Systems Engineering

Human-led design. AI-guided modeling.

Seed Round • Confidential • April 2026

THE PROBLEM

MBSE tools are stuck in the last decade.

Expensive

$5K–$15K+ per seat per year for enterprise MBSE suites. Incumbent pricing locks out 80% of engineering teams.

Desktop-First

No real-time collaboration. No web access. Install on every machine. Disconnected from the rest of the toolchain.

Zero Intelligence

No AI assistance. No ambient suggestions. Engineers manually trace, validate, and cross-reference everything.

Manual tracing takes **5–10 minutes per requirement** — in programs with thousands of requirements, engineers spend more time on traceability than design. (NASA/SERC 2025)

MARKET OPPORTUNITY

$4.5B market.
Growing 12–16% CAGR.

TAM

$4.5B

Composite: MBSE + Requirements + Safety + AI

SAM

$1.8B

Web-native, English-speaking markets

SOM (YEAR 5)

$30–70M

Category leadership, platform revenue

$1.8B MBSE Tools	**$1.2B** Requirements Mgmt
$0.6B Safety & V&V	**$0.5B** AI for Engineering
$0.4B Integration/PLM	

Why now: SysML v2 creates a once-in-a-decade tool replacement window • DoD digital engineering mandates • Local LLMs make on-prem AI viable

THE SOLUTION

The Figma of systems engineering, with an AI partner built in.

Luvian is a web-native MBSE platform where modeling, requirements, safety analysis, and verification live in one connected graph — with ambient AI that learns from your engineering decisions.

🏗️

SysML v2 Native

First-class SysML v2 parser, importer, and visual modeling canvas.

🤖

Ambient ML

6-stage local-first suggestion engine. AI that never sees your classified data.

👥

Real-Time Collab

Yjs/CRDT-powered multiplayer editing. See your team model in real time.

🔒

Deploy Anywhere

Cloud SaaS, on-premises, or air-gapped. One codebase, dual deployment profiles.

PRODUCT DEPTH

One platform. Full coverage.

1 Visual Modeling Canvas

- ✓ Block Definition & Internal Block Diagrams
- ✓ Ports, connectors, nested blocks with auto-layout
- ✓ SysML v2 import & export
- ✓ Interactive canvas with keyboard shortcuts

2 Requirements Engineering

- ✓ INCOSE quality engine (A–F grading)
- ✓ Traceability matrix with coverage analysis
- ✓ Suspect link detection on model changes
- ✓ StrictDoc / ReqIF import

3 Safety Analysis

- ✓ HARA with ISO 26262 ASIL derivation
- ✓ TARA (ISO 21434), STPA, SOTIF, FMEA
- ✓ Hazard-to-requirement linking
- ✓ AI-assisted hazard identification

4 Verification & Validation

- ✓ Test management with evidence attachment
- ✓ Integration readiness reviews
- ✓ Model health validation
- ✓ Coverage engine (7-dimension RCS)

WHY LUVIAN WINS

Technical moat.

Ambient ML Pipeline

6-stage local-first suggestion engine: candidate generation, feature extraction, ranking, policy, response shaping, feedback loop. Runs entirely on-device — no data ever leaves the deployment.

Patent-pending

Command → Patch → CRDT

Dispatch pipeline produces forward + inverse patches that flow into Y.Doc for real-time collaboration with undo/redo preservation. Every operation is reversible.

Patent-pending

Dual Deployment Profiles

Commercial (cloud LLM) and government (local-only AI) profiles from a single codebase. Feature flags control the AI boundary policy.

Air-gap ready

Integrated Decision Graph

Architecture, requirements, safety goals, and test evidence in one connected graph. Change a requirement → suspect links propagate to tests, blocks, and hazards.

Single graph, not bolted-on modules

BUSINESS MODEL

SaaS + on-prem. 4 tiers.

Tier	Target	Price / seat / yr	Key Includes
Team	Small teams (5–20)	**$1,200** ($100/mo)	SysML v2 modeling, collab, requirements, 5 projects
Professional	Mid-market (20–100)	**$2,400** ($200/mo)	+ Safety analysis, INCOSE quality, ambient ML, API
Enterprise	Large programs (100+)	**$3,600** ($300/mo)	+ On-prem, SSO/SAML, audit logs, SLA, dedicated support
Defense/Gov	DoD, air-gap (50+)	**$5,400** ($450/mo)	+ Air-gap AI, IL4/IL5 docs, FedRAMP artifacts, dedicated CSM

70–75%
Subscription Revenue

15–20%
Professional Services

5–8%
AI Compute Add-On

2–5%
Platform / API

3–10x cheaper than incumbent enterprise suites (Cameo, DOORS, Jama). Price-competitive with the new wave at $100–$300/editor/mo — differentiation is depth of integration.

GO-TO-MARKET

Land in engineering. Expand enterprise.

Y1

Design Partners

Founder-led sales. 5 partners. Subsidized pricing for feedback + case studies.

Y2

Early Revenue

First AE. Inbound from content + community. 19 customers. SOC 2.

Y3

Growth

Sales team. Channel partners (Booz Allen, SAIC). FedRAMP. 46 customers.

Y4-5

Scale

Full sales org. Defense enterprise. International. 131 customers.

Target Verticals

- Gov Robotics (DoD Autonomous Systems)
- Software-Defined Vehicles (SDVs)
- Hardware T&V Platforms (dSPACE, Vector, NI)

COMPETITIVE LANDSCAPE

Deepest integration + local-first AI.

Scope vs. AI Sophistication

Capability Coverage

Capability	Luvian	Flow	Cameo	Trace	Dalus
SysML v2 Modeling	✓	✕	✓	✕	✓
Requirements Mgmt	✓	✓	✕	✓	✓
Safety Analysis	✓	✕	✕	✕	~
V&V / Test Mgmt	✓	~	✕	~	~
Ambient / Local AI	✓	~	~	~	~
Real-Time Collab	✓	✓	✕	✓	✓
Air-Gap / On-Prem	✓	✕	✓	✓	~

Only Luvian integrates all four pillars in one connected graph with ambient local-first AI and CRDT-based real-time collaboration.

DEFENSE & GOVERNMENT

Built for classified.
Deployable anywhere.

Air-Gap Ready

Single `docker compose up` deploys the full stack. Local-only AI via 80MB MiniLM embeddings. Zero data egress.

Docker | ONNX Runtime | Local LLM

Standards Literacy

DO-178C, DO-254, ARP4754A/4761, ISO 26262, ISO 21434 built into safety analysis and V&V workflows.

Avionics | Automotive | Defense

Compliance Path

SOC 2 Type 1 targeted post-seed. FedRAMP Li-SaaS planned Y3. IL4/IL5 authorization path mapped for Y4.

SOC 2 | FedRAMP | IL4/IL5

$270K+	$5M+	$15M+
Min defense ACV (50 seats)	Enterprise-wide deal potential	Max defense ACV (500+ seats)

TEAM

Built by engineers who live in the domain.

F

Founder & CEO

Full-stack architect. Built the 3-layer rendering pipeline, ambient ML engine, and defense deployment architecture. Domain expertise in systems engineering tooling.

SW

SW Engineer

20-year veteran. Production hardening: CI/CD, security, observability, performance. Enterprise infrastructure at scale.

PM

Product Manager

Product launch and market fit. Personas, GTM strategy, analytics, onboarding, and beta program execution.

Post-Funding Hiring Plan (Y1)

- Head of Sales
- Solutions Engineer
- Senior Frontend
- Backend / ML Engineer
- Ops / G&A

FINANCIAL PROJECTIONS

Path to $21M revenue by Year 5.

Revenue & Costs (Base Case)

Revenue Scenarios

Base Case	Y1	Y2	Y3	Y4	Y5
Revenue	$165K	$846K	$3.3M	$9.9M	**$21.2M**
Costs	$1.7M	$3.7M	$7.2M	$11.8M	$16.8M
Net Income	($1.6M)	($2.9M)	($3.9M)	($1.9M)	**$4.4M**

UNIT ECONOMICS

Best-in-class LTV:CAC.

LTV:CAC Ratio by Segment (target: >3x)

Gross Margin (blended) **75–80%**

NRR (Y5 target) **130%**

CAC Payback **3–5 mo**

Rev / Employee (Y5) **$299K**

Benchmark: $300–500K for best-in-class SaaS

THE ASK

Raising $7–10M Seed

24+ months runway to product-market fit and first $1M ARR.

Build the Team

4 eng, 1 product, 1 SE, 1 sales, 1 ops. Hire the team that scales past the founder.

Close First Revenue

5 design partners. SOC 2 Type 1. First enterprise contracts. $100K+ ARR by month 12.

Establish the Category

Define "AI-native MBSE" before incumbents respond. File patents. Build advisory board.

Funding Milestones

Seed (now)	Series A (Y2)	Series B (Y3–4)
$7–10M	**$20–30M**	**$50–80M**
PMF signal: 3+ paying, NPS > 40	$1M+ ARR, 120%+ NRR	$5M+ ARR, defense contracts

Luvian

Human-led design. AI-guided modeling.

The platform for safety-critical systems engineering.

Market	$4.5B TAM, 15–16% CAGR
Model	$100–$450/seat/mo SaaS + on-prem
Raise	$7–10M seed, 24+ months runway
Moat	Integrated decision graph + local-first AI

Luvian Confidential — NDA Restrictions